

Expense Tracker – Angular SPA



An Interactive Expense Management System Using Angular

By

Gowtham M

TechMahindra Batch 1

About the Application

- An **Angular Single Page Application (SPA)** for managing expenses with friends
- Allows users to **log in, view friends, track expenses, and manage payments** efficiently.
- Objective :
 - Simplify financial tracking among friends.
 - Provide a **user-friendly** and **interactive interface** for expense management.
 - Ensure seamless **navigation** and **real-time data updates** using Angular services and API calls.

Key Features

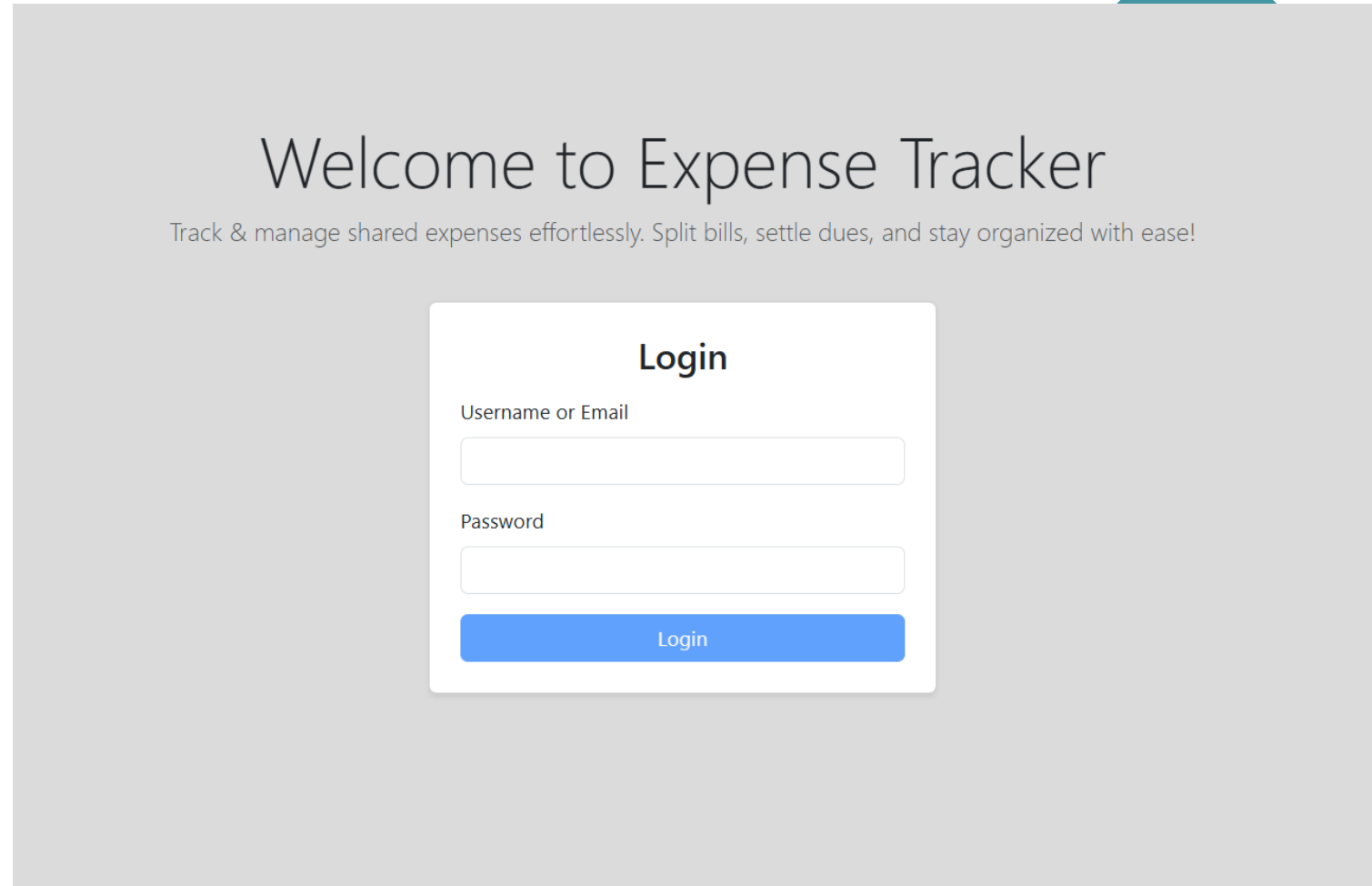
- **Single Page Application (SPA)** – Fast navigation without reloading.
- **Component-Based Architecture** – Reusable and modular components.
- **Parent-Child Communication** – Uses @Input and @Output for data sharing
- **Forms Handling** –
 - Login form(Template-Driven)
 - Expense form(Reactive)
- **Expense Tracking** – Shows amounts owed and received dynamically.
- **Mock API with JSON Server** – Simulates backend data with **db.json**
- **Pipes & Directives** – Used for **formatting** and **UI enhancements**.

Application Workflow :

Screenshot :

1) User Login

- User enters credentials in the **Login Form** (Template-driven-form).
- Authenticated using **Auth Service**.

A screenshot of a web application's login page. The background is a light gray. At the top, the text "Welcome to Expense Tracker" is displayed in a large, dark font. Below it, a smaller line of text reads "Track & manage shared expenses effortlessly. Split bills, settle dues, and stay organized with ease!". In the center, there is a white rectangular box with a subtle shadow. Inside this box, the word "Login" is centered at the top. Below it, there are two input fields: the first is labeled "Username or Email" and the second is labeled "Password". Both fields are empty. At the bottom of the white box, there is a blue button with the word "Login" in white text.

Welcome to Expense Tracker

Track & manage shared expenses effortlessly. Split bills, settle dues, and stay organized with ease!

Login

Username or Email

Password

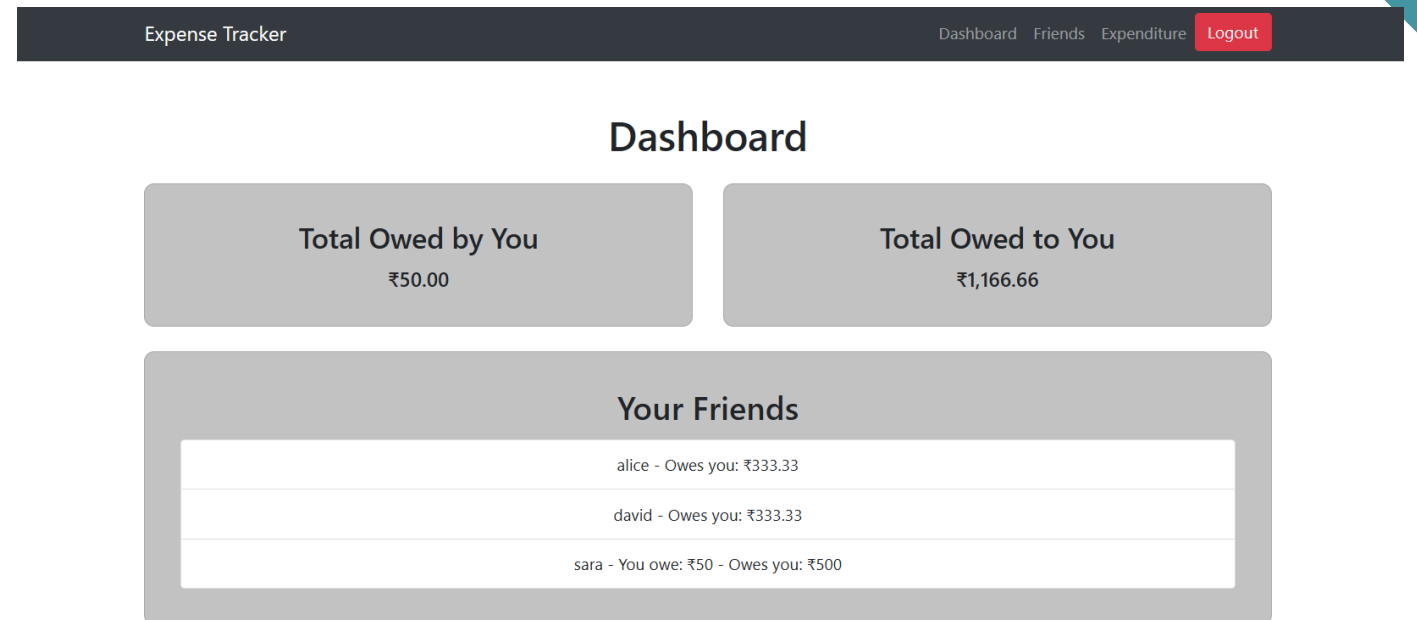
Login

Application Workflow :

Screenshot :

2) Dashboard

- Shown after **successful login**.
- Displays **expense summary**.
- Quick access to **friends & expenditures**.

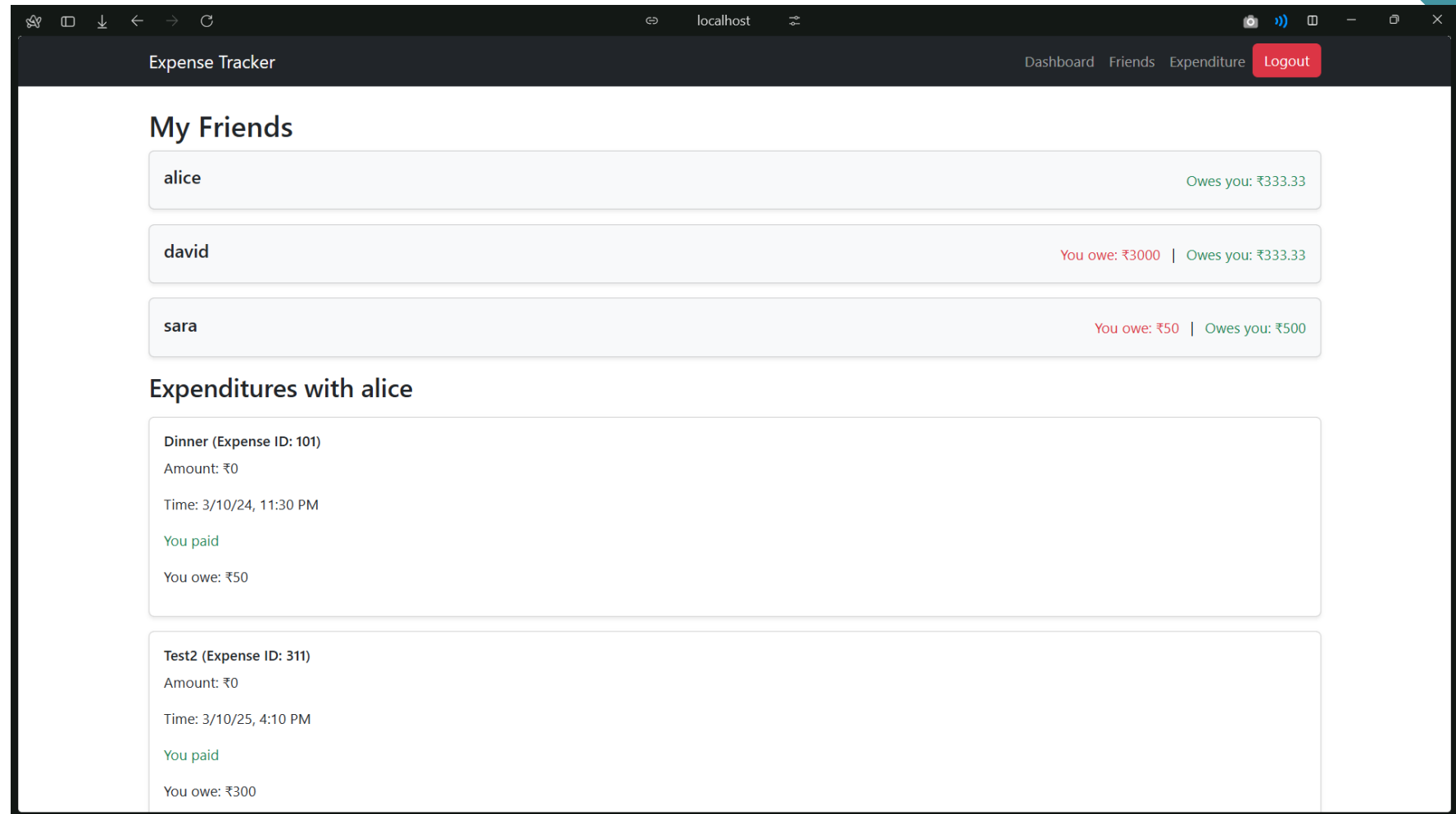


Application Workflow :

3) Friends Page

- Lists all friends with owed amounts.
- Users can view **detailed transactions** by selecting a friend.

Screenshot :



Application Workflow :

4) Expenditure Page

- Users can **add new expenses**.
- Uses **Reactive Form** for input.
- Data stored via **JSON Server (Mock API)**.

Screenshot :

The screenshot displays the 'Expense Tracker' application interface. At the top, a dark navigation bar contains the title 'Expense Tracker' and links for 'Dashboard', 'Friends', 'Expenditure', and a red 'Logout' button. The main content area features a form titled 'Add Expenditure' with the following fields:

- Expenditure Name:** A text input field with the placeholder 'Enter expenditure name'.
- Total Cost:** A text input field containing the value '0'.
- Who Paid?:** A dropdown menu showing 'john' with a downward arrow.
- Split Among:** A text area containing a list of names: 'john', 'alice', 'david', and 'sara'.
- Split Type:** Two radio buttons labeled 'Equally' and 'Manually', with 'Equally' selected.

At the bottom of the form is a dark button labeled 'Add Expenditure'.

Application Workflow :

5) Data Handling & UpdatesUsers

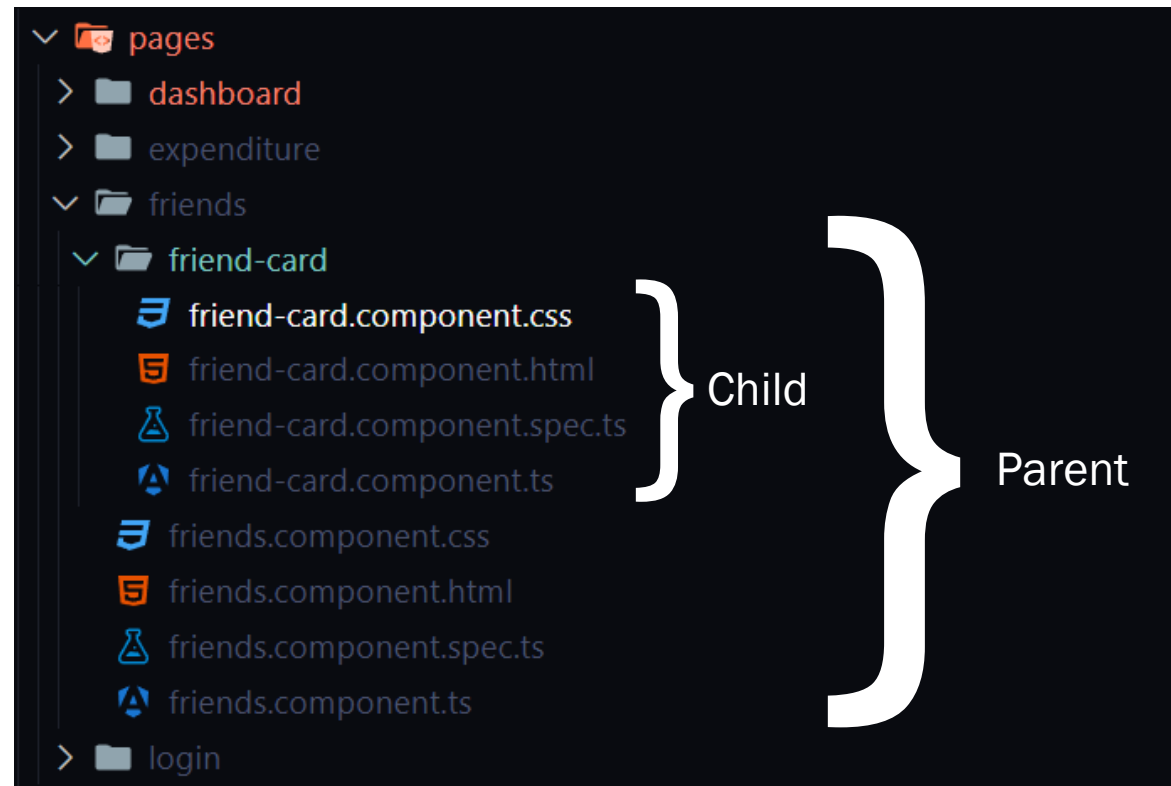
- **Observables** fetch and update data (Both in UI and Json Server)
- Data Handling & Updates

Screenshot :

```
3      "expenditures": [  
4      {  
5          "id": 411,  
6          "userId": 1,  
7          "groupId": 1,  
8          "amount": 1000,  
9          "description": "Car Wash",  
10         "splitAmong": [  
11             1,  
12             4  
13         ],  
14         "owedDetails": [  
15             {  
16                 "payerId": 1,  
17                 "owedBy": 1,  
18                 "amount": 500  
19             },  
20             {  
21                 "payerId": 1,  
22                 "owedBy": 4,  
23                 "amount": 500  
24             }  
25         ],  
26         "timestamp": "2025-03-12T06:09:47.904Z"  
27     }  
28 ]  
29 }
```


Angular Concepts Used :

- **Components** – Modular design with **Dashboard**, **Friends**, **Expenditure**, **Navbar**, and **Login** components.
- **Parent-Child Components** – Data sharing between **FriendsPage** & **Friends Card**



Angular Concepts Used :

- **Routing** – Navigation between **Dashboard, Friends, and Expenditure** using Angular Router
- **Pipes** – Used for **formatting currency and filtering data**.
- **Directives** :
 - Structural Directives(`*ngIf` , `*ngFor`) for dynamic content.
 - Attribute Directives (`[ngClass]` , `[ngStyle]`) for styling.
- **Forms**
 - **Template-driven Form** for Login.
 - **Reactive Form** for adding expenses.
- **Services & Observables** :
 - **AuthService** for authentication.
 - **FriendsService** for fetching and updating data.

Technologies Used

- **Frontend:** Built with **Angular & TypeScript**, using **HTML & CSS** for structuring and styling.
- **Backend (Mock API):** Uses **JSON Server** to simulate a backend for managing user data and expenses.
- **State Management:** Implements **Angular Services** and **RxJS Observables** to handle data flow efficiently.
- **Routing:** **Angular Router** enables smooth navigation between pages.
- **Forms:** Utilizes **Template-Driven and Reactive Forms** for handling user inputs in login and expense entry.
- **Styling:** Designed with **CSS** and **Bootstrap** for responsive UI.



Thank you

Gowtham M

gowthamudpm2004@gmail.com

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